

B04 Gate B Protected Review - Inventory Replenishment

Protected diagnostic review - fresh evidence and changed retry

Protected review - saved genuine Gate attempt required

Open only after the matching Gate attempt is saved. Diagnose the smallest weak competency, repair it, close this review, then save changed evidence. Gate B/C remain freshness-protected.

Objective source controls

warehouses 8; products 30; supplier changes 36; inventory snapshots 720; purchase orders 48; receipts 37; reorder rules 120

Metric	Protected control
All snapshot on-hand sum	41275
Latest snapshot date on-hand	4105
Snapshot dates	10 daily dates, 72 rows each
Orphan receipt	PO 99999 / received_qty 25
Conflicting supplier history	P012 @ 2026-04-01 (SUP01 vs SUP07)
Closed purchase orders	36
Matched receipt quantity	1404

Why plausible output can still be wrong

- Inventory on hand is a snapshot measure: summing 10 days to 41275 is not a valid current-stock total; the latest-day total is 4105.
- P012 has two suppliers on the same effective date, so as-of supplier history is ambiguous until repaired.
- The orphan receipt must not be deleted to make referential checks pass.
- Reorder rules are product-warehouse grain; do not assume every possible product/warehouse pair has a rule.

Reasoning and evidence rubric

- State source grain and key before designing joins or layers.
- Separate current master from effective-dated history and reject overlapping/conflicting intervals.
- Name durable raw evidence and the transformation/promotion boundaries.
- Define block, warn and reconcile rules with owners and evidence.
- Predict fan-out/semi-additive risk before aggregating.
- Keep orphans and rejected rows in an auditable exception path.
- Reconcile row counts and business totals independently before trusted promotion.
- For backfill/replay, bound the slice, preserve raw evidence, prevent duplicate publish and prove reconciliation after rerun.

Critical failures

- Hidden data loss or deleted exception evidence.
- Ambiguous history accepted as truth.
- Snapshot or many-side joins produce inflated but plausible measures.
- Higher-grain targets repeated at transaction grain.
- Unbounded replay or unsafe state advancement.
- No independent reconciliation or ownership.

Repair and changed retry route

- Keep the original Gate attempt append-only.
- Open only this matching protected review after the saved genuine attempt.
- Repair the named weakness using the targeted remediation cards.
- Close the review and perform a changed micro-proof in a different small scenario.
- Save the retry evidence separately. Gate B/C exposure still requires the existing explicit reassignment controls.

Next route

PASS after explicitly assigned full reassessment -> B05. If instability remains, repair; Gate C stays sealed.